

38

CrewAI

Teams of AI agents working together

Many agents, each with a role, like a real team

[AI/ML Engineer Learning Series · Deep Dive Edition](#)

What it is	A framework for multi-agent teams
Core idea	Several agents, each with a role, collaborating
Like	A real team: researcher, writer, editor
Best for	Complex tasks split across specialists
Builds on	The agent concepts you just learned

What's Inside

Here is everything covered in this guide, in order:

#	Section	What you'll learn
1	What is CrewAI?	Teams of agents
2	Why Multiple Agents?	Specialists beat a generalist
3	The Crew Idea	Roles, tasks, and a goal
4	The Building Blocks	Agent, task, crew, process
5	Defining Agents	Giving each a role
6	Defining Tasks	The jobs to be done
7	Running a Crew	Putting it together
8	How Agents Work Together	Sequential vs hierarchical
9	Common Mistakes	What to avoid
★	Cheatsheet	Quick reference

1. What is CrewAI?

CrewAI is a framework for building teams of AI agents that work together. Where PydanticAI focuses on one reliable agent, CrewAI focuses on many agents collaborating — each with its own role, skills, and job, like a real team of people. The name says it: you build a 'crew' of agents.

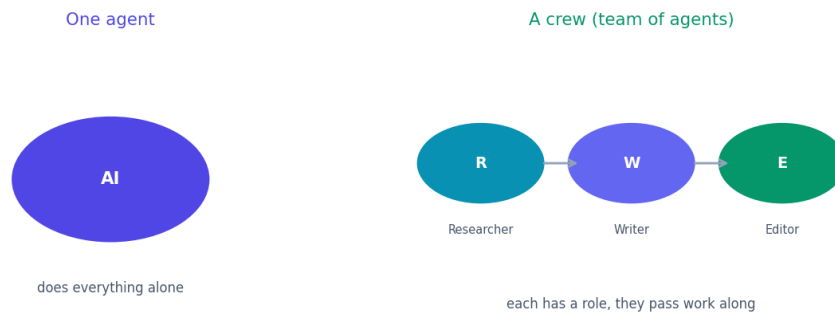


Fig 1 — One agent does everything alone. A crew splits the work across specialists who pass work along.

Imagine writing a research report. One agent doing it all might be okay. But a team — a researcher to gather facts, a writer to draft it, and an editor to polish it — produces better work, just like with people. CrewAI lets you build exactly that kind of team of cooperating agents.

■ *CrewAI builds directly on the agent ideas from Topic 36. Each crew member is an agent with the Think-Act-Observe loop and tools. CrewAI just coordinates several of them into a team.*

2. Why Multiple Agents?

Why not just one powerful agent? Because splitting a big job across specialists often works better — the same reason human teams exist. Each agent can focus on one thing and do it well, with its own role, instructions, and tools.

Benefit	Why it helps
Specialisation	Each agent is expert at one job
Focus	Simpler, clearer instructions per agent
Better quality	Work is reviewed and refined by others
Easier to manage	Break a big task into clear roles
Reusable	Mix and match agents for new tasks

A single agent trying to do everything can get confused or cut corners. A crew divides the work so each agent has a clear, narrow job — leading to more reliable, higher-quality results on complex tasks.

3. The Crew Idea

A crew is a group of agents, each assigned tasks, all working toward a shared goal. Work flows from one agent to the next, just like a real team passing a project along.



Fig 2 — A crew in action: a researcher gathers info, passes it to a writer, who passes a draft to an editor.

In this example, the researcher agent finds the facts and hands them to the writer agent, who drafts the content and hands it to the editor agent, who polishes the final result. Each agent does its part and passes the work forward. The output of one becomes the input of the next — a smooth pipeline of specialists.

4. The Building Blocks

CrewAI has a few simple building blocks. Once you know these four, you can build any crew.



Fig 3 — The CrewAI building blocks: agents, tasks, tools, the crew, and the process.

Block	What it is
Agent	An AI worker with a role, goal, and background
Task	A specific job, assigned to an agent
Tools	Abilities agents can use (search, your RAG, APIs)
Crew	The whole team of agents and their tasks
Process	How the crew works (sequential or hierarchical)

You define agents (the workers), give them tasks (the jobs), optionally hand them tools, then gather them into a crew with a process that says how they cooperate. That's the whole model.

5. Defining Agents

Each agent gets a role, a goal, and a backstory. The backstory is a short description that shapes how the agent behaves — it's prompt engineering (Topic 30) giving each agent its personality and expertise.

```
# pip install crewai
from crewai import Agent

researcher = Agent(
    role='Senior Researcher',
    goal='Find accurate, up-to-date facts on the topic',
    backstory='You are an expert researcher who digs deep '
              'and never settles for shallow answers.',
    tools=[search_tool]      # give it tools
)

writer = Agent(
    role='Tech Writer',
    goal='Write a clear, engaging article from the research',
    backstory='You turn complex facts into simple, '
              'reader-friendly writing.'
)
```

■ *The role, goal, and backstory are how you steer each agent. Clear, specific roles make a big difference — vague roles lead to agents that overlap or get confused. This is your prompt engineering skill applied per agent.*

6. Defining Tasks

A task is a specific job assigned to an agent, with a description of what to do and what the result should look like. Tasks are how you tell each agent exactly what you need from them.

```
from crewai import Task

research_task = Task(
    description='Research the latest trends in AI agents.',
    expected_output='A list of 5 key trends with brief notes.',
    agent=researcher      # who does this task
)

writing_task = Task(
    description='Write a short article using the research.',
    expected_output='A 300-word article.',
    agent=writer
)
```

Notice `expected_output` — it tells the agent the shape of result you want, just like good prompting. And each task is tied to the agent best suited for it. The writer's task naturally uses the researcher's output as its input.

7. Running a Crew

Finally, you gather the agents and tasks into a crew and run it. CrewAI handles the coordination — passing work between agents until the goal is complete.

```
from crewai import Crew, Process

crew = Crew(
    agents=[researcher, writer],
    tasks=[research_task, writing_task],
    process=Process.sequential # do tasks in order
)

# Run the whole crew
result = crew.kickoff()
print(result)
```

That's a complete multi-agent system. `crew.kickoff()` starts the team: the researcher does its task, its output flows to the writer, and you get the final article. CrewAI manages the hand-offs so you don't have to wire them manually.

■ *This is the whole pattern: define agents, define tasks, build the crew, kickoff. Start with two agents and a simple sequential process — you can always add more specialists later.*

8. How Agents Work Together

CrewAI offers two main ways for a crew to operate, called processes. They decide how work moves through the team.

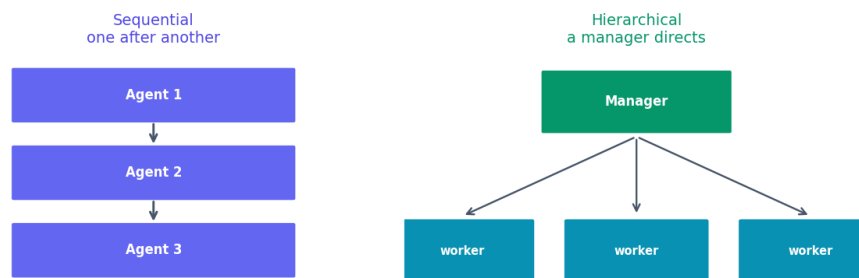


Fig 4 — Sequential: agents work one after another. Hierarchical: a manager agent directs the others.

Process	How it works
Sequential	Agents work one after another, in order — simplest
Hierarchical	A manager agent assigns and reviews the others' work

Sequential is the easy starting point: task 1, then task 2, then task 3, each feeding the next. **Hierarchical** adds a manager agent that delegates tasks to workers and checks their results — more like a real company structure, useful for complex projects. Start sequential; move to hierarchical when you need

coordination.

9. Common Mistakes

Mistake	Why it's a problem	Fix
Vague agent roles	Agents overlap or get lost	Give clear, distinct roles
Too many agents	Slow, confusing, costly	Start with 2-3 agents
No expected_output	Fuzzy task results	Describe the output you want
Wrong process choice	Bad coordination	Start sequential, then scale up
Ignoring cost	Many agents = many LLM calls	Watch usage; keep crews small
One agent could do it	Needless complexity	Use a crew only for real teamwork

Quick Reference Cheatsheet

Concept / Task	Meaning or Code
CrewAI	framework for teams of agents
Crew	a team of agents + tasks
Agent	a worker with role, goal, backstory
Task	a job assigned to an agent
Process	sequential or hierarchical
Install	<code>pip install crewai</code>
Make agent	<code>Agent(role=, goal=, backstory=)</code>
Give tools	<code>Agent(..., tools=[...])</code>
Make task	<code>Task(description=, expected_output=, agent=)</code>
Build crew	<code>Crew(agents=[...], tasks=[...], process=)</code>
Run	<code>crew.kickoff()</code>
Best for	complex tasks split across specialists

The one idea to remember: CrewAI builds teams of AI agents, each with a role and tasks, working together toward a goal — like a researcher, writer, and editor. Splitting a complex job across focused specialists produces better results than one agent doing everything.

Next Topic: Streamlit — the fastest way to turn your Python AI projects into interactive web apps that anyone can use, with no web development needed. The start of the deployment stage.